

within the speaker unit using the inbuilt sound chip, and thus is not worth it if you're using an expensive sound card. But we have to admit that it sounds a lot better than most onboard sound solutions we've heard. Apple Macs are not officially supported.

Digital connectors of any kind were non-existent in the speakers we tested. A digital connector ensures a pure audio signal, devoid of noise of any kind, to deliver the highest-quality input to the speakers. Auxillary inputs were there on all the speakers except the Altec Lansing, Creatives, XFreer, Intexes, all the Tech-Coms (except the SSD-5101R and SSD-6001R), and the Zebronics ZEB-SW6900R. An auxillary input lets you connect a stereo sound source such as a CD player or hi-fi music system to the speakers without having to remove an already-connected 5.1 source. The Tech-Com SSD-5101R and Artis S8800 even had dual auxillary inputs, allowing you to connect a second stereo source.

Accessories

An external decoder decodes audio signals from the sound source if the source contains information complying with audio digital standards such as Dolby Digital or DTS. This is pure digital audio and is preferred by audiophiles.



Unfortunately, none of these speakers came with an external decoder.

With the exception of the Creative SBS 580, all the speakers came with either a wired or a wireless remote control. One of the most stylish remotes came with the Altec Lansing FX5051. The wired remote has a smooth control knob and blue LEDs, and the slick, angled wireless remote was the icing on the cake.

Radio functionality is usually not seen on expensive speaker sets, and is very rare in the case of 5.1 sets. Only the Artis S6600R/FM came with this functionality. We were very pleased to see that not only did it have FM radio, but also AM radio so you can play your favourite medium-wave regional channels.

5.1 speaker sets are always accompanied with a huge bundle of cables. We would like to note here that most of the Artis speakers came with colour-coded cables, which eases the effort required to connect different sound channel pairs from the sound card to the correct inputs on the speaker sets / amps. Bare-ended wires were also bundled with quite a few of the speaker sets that featured clamp type connectors. While some of these cables were of inferior quality, such as those which came with the Tech-Com SSD-5101R, the ones that accompanied the Artis sets, espe-

Expert Tests

For this speaker test, we decided to call a professional with a trained ear and some testing equipment to give us an idea of how good multimedia speakers are from the perspective of an audiophile. This was done to gauge whether professional testing equipment corroborated what our ears heard.

We asked Sunil Karanjikar from SOUND.COM (www.online-sound.com), a professional live sound provider, to come in and take a look at some of the top-performing sets in the 2.1 and 5.1 categories.

Sunil is a professional live sound engineer, with over six years of experience. Although he specialises in live sound and concerts, he is an audiophile and designs his own speakers in his spare time.

Test Equipment

SIA SmaartLive 5 FFT based RTA spectrum analyser: SIA's dual-channel FFT-based audio measurement software is the professional audio industry standard for real-time sound system measurement, optimisation, and control. It combines powerful audio frequency measurement, analysis and data logging capability with a highly intuitive, user-friendly interface and the ability to remotely control an extensive list of top name equalizers and DSP processors. This was used to compare the measured response with a reference signal. (Details: http://www.siasoft.com/products/smaart_live.html)

Presonus Firebox Audio Interface: The Firebox is a professional grade sound card that connects via Firewire to bypass a system's soundcard. It offers much better frequency response compared to consumer level audio solutions. (Details: <http://www.presonus.com/firebox.html>)

Audix TR40 Measurement Microphone: This is a flat frequency response microphone that is specially used for testing and measurement purposes. (Details: www.audixusa.com, found under Contractor Microphones)

Neutrik Minirator Test Tone Generator: This handy portable gadget generates all kinds of test tones—pink noise, white noise, sine / square waves, sweep signals (20Hz to 20 kHz).



Sunil Karanjikar at a live event

Mackie VLZ Pro 1402: A professional 14-channel mixer that was used to route the sound from the Minirator to the speakers and the Firebox. (Details: <http://www.mackie.com/products/1402vlzpro/>)
HP Pavilion dv2119TX Notebook: This laptop has Intel's Core 2 Duo CPU and 1 GB of DDR 2 RAM.

Test Procedure

The testing was done by generating "pink noise" and sending it to the speakers via the mixer. Pink noise is acoustical energy distributed uniformly by octave throughout the audio spectrum (the range of human hearing, approximately 20 Hz to 20 kHz). Most people perceive pink noise as having uniform spectral power density—the same apparent loudness at all frequencies. In pink noise, the total sound power in each octave is the same as the total sound power in the octave immediately above or below it. In all the graphs displayed, pink noise is shown in light blue.

All the speakers were set to 75 per cent of their maximum volume, so as to get their ideal output without distortion.

The output of the speaker sets were recorded by the Audix TR40 Measurement Microphone and then fed back to the laptop and analysed with the SmaartLive software.

There are two waveforms displayed in the software—one of the reference signal (light blue wave) that was sent to the speakers, and the other is the measured signal that the speakers sent out. By comparing the two waveforms, it was possible to gauge how well or poorly the speaker set was able to reproduce the sound it was supposed to.

It must be noted that the acoustic signal takes time to reach the microphone and the software has to compensate for this delay. It automatically measures the time difference between the reference and measured signals and corrects it by adding the delay to the reference signal. This is displayed in both feet and milliseconds, such as 7.80 feet/6.92 ms.